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19EAC384 – Computer Networks Lab (S6 EAC)

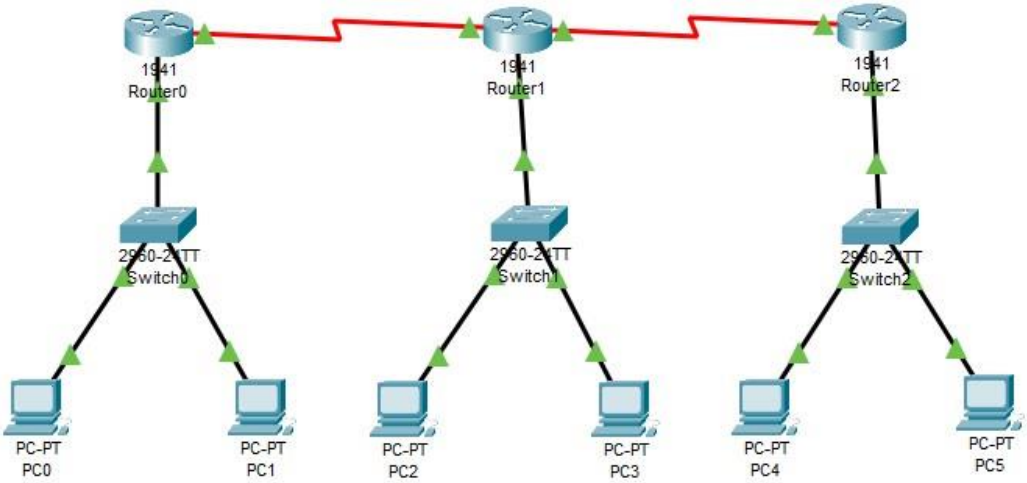
LABSHEET 5 - Assignment

STATIC ROUTING

Aim:

- To implement static routing by configuring multiple routers in a network. In this exercise, the student will learn how to manually specify the configuration of the routers.

Exercises-1: Configure a WAN set up using three routers, three switches and 6 PCs to implement the concept of Static Routing using Cisco Packet Tracer. The following steps may be followed:



Configuration For Router0:

```
R0(config)#
R0(config)#interface GigabitEthernet0/0
R0(config-if)#ip address 220.58.1.1 255.255.255.0
R0(config-if)#no shutdown
R0(config-if)#
R0(config-if)#interface Serial0/0/0
R0(config-if)#ip address 10.0.0.1 255.255.255.252
R0(config-if)#no shutdown
R0(config-if)#
R0(config-if)#ip route 220.58.2.0 255.255.255.0 10.0.0.2
R0(config)#ip route 220.58.3.0 255.255.255.0 10.0.0.2
R0(config)#
R0(config)#end
R0#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R0(config)#
R0(config)#
%SYS-5-CONFIG_I: Configured from console by console
```

Configuration For Router1:

```
R1(config)#
R1(config)#interface GigabitEthernet0/0
R1(config-if)#ip address 220.58.2.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#interface Serial0/0/0
R1(config-if)#ip address 10.0.0.2 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#interface Serial0/0/1
R1(config-if)#ip address 10.0.0.5 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#ip route 220.58.1.0 255.255.255.0 10.0.0.1
R1(config)#ip route 220.58.3.0 255.255.255.0 10.0.0.6
R1(config)#
R1(config)#end
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z
R1(config)#
R1(config)#
%SYS-5-CONFIG_I: Configured from console by console
```

Configuration For Router2:

```
R2(config)#
R2(config)#interface GigabitEthernet0/0
R2(config-if)#ip address 220.58.3.1 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#
R2(config-if)#interface Serial0/0/0
R2(config-if)#ip address 10.0.0.6 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#
R2(config-if)#ip route 220.58.1.0 255.255.255.0 10.0.0.5
R2(config)#ip route 220.58.2.0 255.255.255.0 10.0.0.5
R2(config)#
R2(config)#end
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#
R2(config)#
%SYS-5-CONFIG I: Configured from console by console
```

Ping Commands:

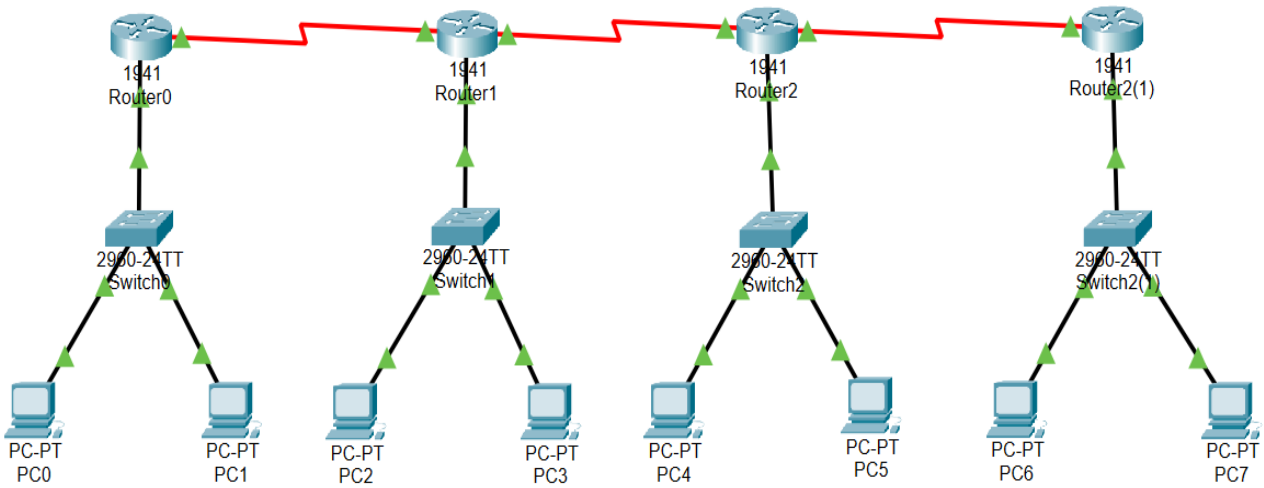
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 220.58.1.2

Pinging 220.58.1.2 with 32 bytes of data:

Reply from 220.58.1.2: bytes=32 time<1ms TTL=128
Reply from 220.58.1.2: bytes=32 time<1ms TTL=128
Reply from 220.58.1.2: bytes=32 time<1ms TTL=128
Reply from 220.58.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 220.58.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Assignment: Attempt the same steps as above to configure static routing for 4 routers interconnecting LANs.




```
R2(config)#
R2(config)#interface GigabitEthernet0/0
R2(config-if)#ip address 220.58.4.1 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#
R2(config-if)#interface Serial0/0/0
R2(config-if)#ip address 10.0.0.6 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#clock rate 64000
This command applies only to DCE interfaces
R2(config-if)#
R2(config-if)#exit
R2(config)#
R2(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.5
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 220.58.4.2

Pinging 220.58.4.2 with 32 bytes of data:

Reply from 220.58.4.2: bytes=32 time=4ms TTL=128
Reply from 220.58.4.2: bytes=32 time=4ms TTL=128
Reply from 220.58.4.2: bytes=32 time=4ms TTL=128
Reply from 220.58.4.2: bytes=32 time=4ms TTL=128

Ping statistics for 220.58.4.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 4ms, Average = 4ms
```

Inference:

While setting up static routing between four routers connected to different LANs, I realized how important it is to plan IP addresses correctly and configure routes properly for networks to communicate. At first, I struggled with choosing the right interfaces and assigning the correct IP addresses, but fixing those issues helped me better understand how routers connect. This activity gave me hands-on experience in setting static routes to allow data to move between networks and improved my skills in using the command line to set up WAN connections.